# Perforated Structures as Archive:
# a ceramic artistic research on colonial erasure in Brazilian architecture

Estruturas Vazadas como Arquivo:
uma pesquisa artística em cerâmica sobre apagamento colonial na arquitetura brasileira

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2025

**Abstract**

This article examines the historical, material and symbolic trajectory of perforated architectural structures in Brazil, with particular attention to the muxarabi and the cobogó. Rather than asserting a direct origin between them, the research proposes a critical reading of their formal and functional affinities: both treat the wall as a membrane, filtering light, air, shade and gaze in hot-climate contexts. Drawing on decolonial studies (Quijano, Mignolo), material culture anthropology (Ingold, Appadurai), phenomenology of space (Bachelard, Merleau-Ponty), Brazilian debates on architecture, heritage and coloniality (Moassab and Name; Name and Zambuzzi), and recent research on environmental comfort, the article proposes the concept of coloniality of dwelling to consider how certain constructive solutions have been devalued, erased, or reintroduced under new material and aesthetic languages. The early nineteenth-century removal and devaluation of perforated elements associated with Arab-Islamic and Luso-Brazilian repertoires, together with the later valorisation of the cobogó in Brazilian modern architecture, are analysed not as a linear genealogy, but as signs of a dispute over memory, climate, body and modernity. The research unfolds into an artistic ceramic practice in which perforated structures operate as material archive and poetic argument.

**Resumo**

Este artigo examina a trajetória histórica, material e simbólica de estruturas arquitetônicas vazadas no Brasil, com atenção especial ao muxarabi e ao cobogó. Em vez de afirmar uma origem direta entre ambos, a pesquisa propõe uma leitura crítica de suas afinidades formais e funcionais: ambos operam a parede como membrana, filtrando luz, ar, sombra e olhar em contextos de clima quente. A partir do diálogo entre estudos decoloniais (Quijano, Mignolo), antropologia da cultura material (Ingold, Appadurai), fenomenologia do espaço (Bachelard, Merleau-Ponty), debates brasileiros sobre arquitetura, patrimônio e colonialidade (Moassab e Name; Name e Zambuzzi) e pesquisas recentes sobre conforto ambiental, o artigo propõe o conceito de colonialidade do habitar para pensar como certas soluções construtivas foram desvalorizadas, apagadas ou reintroduzidas sob novas linguagens materiais e estéticas. A remoção e desvalorização de elementos vazados associados a repertórios

árabe-islâmicos e luso-brasileiros no início do século XIX, bem como a posterior valorização do cobogó na arquitetura moderna brasileira, são analisadas não como uma genealogia linear, mas como sinais de uma disputa sobre memória, clima, corpo e modernidade. A pesquisa se desdobra em uma prática artística em cerâmica, na qual estruturas vazadas funcionam como arquivo material e argumento poético.

**Keywords**

cobogó; mashrabiyya; muxarabi; coloniality of dwelling; material culture; ceramics; Brazilian architecture; environmental comfort.

**Palavras-chave**

cobogó; muxarabi; colonialidade do habitar; cultura material; cerâmica; arquitetura brasileira; conforto ambiental.

## 1. Introduction

Architecture is usually read as an expression of its time: styles, techniques and materials that reflect the taste or necessity of a given historical moment. Less frequently is the question asked of what architecture erases: which constructive knowledge was classified as backward, which climate-responsive solutions were replaced by imported models, and what those erasures reveal about the persistence of colonialism beyond the political events that give it a name.

This article departs from a small and concrete object: the cobogó, a perforated block of concrete or ceramic widely used in Brazilian architecture, patented in Recife in 1929. Its destination, however, is a broader question: when a constructive solution associated with the body, climate and popular everyday life is devalued, forgotten and later reactivated as modernity, what does that movement reveal about the still-active mechanisms of colonial erasure? Who, in the end, holds the right to name, to build, to dwell?

The initial hypothesis of this research approached the cobogó and the muxarabi as if there were an almost direct relationship between them. Bibliographic review, however, requires a more careful formulation. The cobogó should not be presented as a direct descendant of the muxarabi. The more defensible relationship is one of formal, climatic and cultural affinity: both belong to a broad family of perforated devices that respond to heat, luminosity, ventilation and privacy through the permeable wall.

This change does not weaken the decolonial dimension of the research. On the contrary, it makes it more precise. The problem is not to prove a single origin, but to observe how certain constructive intelligences were disauthorized when associated with the colonial past, the Arab-Islamic world or popular knowledge, and could return when translated through modern, industrial and national languages. Coloniality of dwelling, the concept proposed in this article, names precisely this silent operation

through which architecture organises hierarchies of visibility, comfort, belonging and memory.

The article is structured as follows: section 2 presents the methodological approach; section 3 reconstructs the critical genealogy of the problem, from the mashrabiyya and the muxarabi to the modern cobogó; section 4 describes the ceramic artistic research developed in parallel with the theoretical work; section 5 discusses the findings, limits and contradictions; section 6 presents concluding remarks.

## 2. Methodological approach

This research is theoretical, historical and artistic in nature. Its method combines bibliographic review, critical reading of secondary sources on Brazilian colonial legislation and architecture, analysis of architectural objects as material documents, and ceramic experimentation. It is not empirical research in the quantitative sense: the object of analysis is the artefact itself - its form, its historical trajectory, the uses it accumulates and the silences it carries.

The methodological choice to begin from the object rather than from a closed hypothesis is deliberate. Taking the cobogó as the entry point, the research operates as a critical excavation of layers of meaning: beneath the modern form lies the question of industry; beneath industry, the reactivation of climatic solutions; beneath this reactivation, the memory of colonial perforated elements; beneath those elements, an Iberian heritage marked by centuries of Arab-Islamic presence. This excavation does not aim to demonstrate a direct line of descent, but to make visible approximations, displacements and interruptions.

Sources consulted include specialist literature on Brazilian architecture, Islamic architecture and environmental comfort; theoretical texts from decolonial, phenomenological and material culture fields; recent research on thermal performance and natural ventilation through cobogós; and contemporary discussions that help situate the renewed visibility of the object. Non-academic materials are not treated here as theoretical authorities, but as indications that simplified origin narratives continue to circulate and therefore require critical scrutiny.

The ceramic artistic research described in section 4 functions as a sensitive extension and test of the theoretical hypotheses. The pieces do not illustrate the text; they formulate the problem in another language. The work is developed between Brazil and Germany, which creates a methodological displacement of its own: looking at Brazilian architecture from the outside shifts what becomes visible. Elements that in Brazil may appear banal reveal themselves, at a distance, as the result of historical choices, material substitutions and symbolic erasures.

## 3. Genealogy of an element and theoretical framework

### 3.1 From mashrabiyya to muxarabi: spatial technology, climate and control

The mashrabiyya is an architectural element associated with different contexts of the medieval and modern Islamic world. In its best-known forms, it appears as a wooden lattice in windows or projecting balconies, filtering light, promoting ventilation and preserving interior privacy. Its etymology is commonly linked to the Arabic verb sharaba, to drink, in reference to the use of water jars near openings and to evaporative cooling. Before being ornament, it is a spatial and climatic technology.

Its spatiality is asymmetric: from inside, the exterior can be seen; from outside, the interior remains partially concealed. This asymmetry organised specific social relations, including gendered ones, in certain Islamic urban contexts. There is a constitutive ambiguity here that this research does not intend to resolve: the muxarabi can be read at once as an intelligent climatic response and as a device for controlling presence. What filters the air also filters the gaze.

The arrival of similar devices in Brazil follows a long and mediated path. Between the eighth and fifteenth centuries, much of the Iberian Peninsula was under Islamic rule, a period during which Arab, Berber, Jewish and Christian cultures produced hybrid architectural traditions. Elements such as rótulas, gelosias and muxarabis arrived in Luso-Brazilian architecture not as pure transplants, but as adaptations of adaptations, incorporated into Portuguese popular architecture and reinterpreted in the climatic conditions of colonial Brazil.

Brazilian literature on the subject records that these elements were widely used from the seventeenth to the early nineteenth century, especially as responses to excessive solar radiation, the need for ventilation, and the separation between public and private space (Vettorazzi et al., 2022). In 1809, by royal decree, the use of elements classified as Oriental was prohibited or discouraged on facades, in a context of approximation to European, Neoclassical and industrial aesthetics. More than an instantaneous disappearance, this episode should be understood as part of a political, aesthetic and material change in the urban image.

For this reason, the near-extinction of muxarabis in Brazil cannot be read as a simple natural evolution of taste. It expresses a historical choice: to replace a popular, climatic and hybrid technology with a visuality considered more civilised, European and modern. At this point, architecture becomes an archive of power.

### 3.2 The cobogó: modern invention, contested continuity

In 1929, in Recife, a constructive piece was patented that would receive the name cobogó - a term formed from the initial syllables of the surnames of Amadeu Oliveira Coimbra, Ernest August Boeckmann and Antônio de Góis. Unlike a direct genealogical reading, which would make the cobogó a simple descendant of the muxarabi, recent research suggests a more complex relationship. The cobogó initially emerged as a prefabricated industrial block, designed to make construction faster, more practical and more economical. Its bioclimatic function - filtering light, creating shade, allowing

ventilation and producing transitional zones - was consolidated through popular use and Brazilian modern architecture.

This distinction matters for two reasons. First, the emergence of the cobogó belongs to a different historical, material and urban conjuncture. Second, its invention should not be reduced to a conscious recovery of the muxarabi. Recent scholarship suggests that its first formulation was tied to industrial prefabrication and constructional practicality, while its aesthetic, climatic and thermal meanings were consolidated through subsequent uses and through its appropriation by architects and residents in hot-climate contexts.

Incorporated by architects such as Luiz Nunes, especially in the Olinda Water Tower in the 1930s, the cobogó became part of the repertoire of Brazilian modern architecture. What matters to this research is not to deny its Brazilian invention, but to ask why a perforated solution could be celebrated as modern when produced through concrete, modularity and industry, while previous elements, also perforated and climatically intelligent, had been associated with backwardness, the colonial or the Oriental.

The cobogó does not need to be understood as a direct continuation of the muxarabi in order to carry an earlier architectural memory. What returns in it is not necessarily a single origin, but a constructive logic recurring in hot-climate architectures: the wall as membrane, not as barrier; the void as passage for air, light, shade and gaze; separation as filter rather than total closure.

In this sense, the relationship between muxarabi and cobogó should be understood less as linear lineage and more as transformed survival. The expression does not designate historical proof of descent, but a critical operation: what was devalued under one aesthetic regime may reappear when translated by another material and symbolic regime.

### 3.3 Theoretical framework: coloniality, material culture, phenomenology

The concept of coloniality of power, elaborated by Aníbal Quijano, refers to the persistence of colonial domination after the end of colonial administrations, operating through economic, racial, cultural and epistemic structures. Walter Mignolo deepens this analysis by showing that Western modernity was constituted through the invisibilisation of other ways of knowing. Recent research in architectural history and heritage has displaced this discussion into the built environment, showing that buildings, techniques and material archives also participate in the production of conflicting and silenced memories (Egbers et al., 2024).

Drawing on these contributions, this article proposes the concept of coloniality of dwelling: the way in which architecture and built space organise, materialise and naturalise colonial hierarchies of belonging, comfort and visibility. The concept does not replace coloniality of power, knowledge or being; it brings them closer to the everyday

scale of the house, the window, the facade, the body that feels heat, and the shade that makes dwelling possible.

To think of the cobogó as material archive, the article draws on Tim Ingold, for whom objects and materials should be understood as processes in flow, and on Arjun Appadurai, who argues that things have social lives, moving through different regimes of value. In this perspective, the cobogó is not only an architectural object: it is a material trajectory that accumulates uses, discourses, forgettings and reactivations.

The phenomenological dimension, articulated through Bachelard and Merleau-Ponty, holds that inhabited space is not neutral geometry but a field of bodily experience. The cobogó only makes sense from a situated body: a body that needs air, shade and ventilation in order to dwell. Recent research on thermal performance reinforces this point by indicating that perforated elements can contribute to natural ventilation, environmental comfort and reduced dependence on mechanical systems in vulnerable housing (Stefano and Figueiredo, 2024). To build with the void, in this reading, is a form of environmental knowledge.

## 4. The artistic research: ceramics as archive

The theoretical research unfolds into a ceramic artistic practice whose results - objects and processes - function as sensitive extensions of the hypotheses elaborated. The pieces work with ceramic lattices that formally approach the cobogó: perforated structures that sustain their own form through the tension of the mesh, not through the compact mass of the block.

The choice of ceramics is not arbitrary. The material carries internal tension: it is earth, weight, resistance, but also fragility. By transforming a flexible lattice into a rigid object, a contradiction is produced that is the central argument of the pieces: the object appears light, but may break; appears functional, but is unstable; appears decorative, but is structural. Following Ingold, the process does not consist in imposing a form on the material, but in corresponding with it - responding to what the clay allows, refuses and suggests.

The openings that structure the pieces - gaps, passages, voids - produce, when light passes through them, shadow projections that vary with time and with the observer's position. In this behaviour, the climatic logic of many perforated structures is reproduced at a domestic scale: shadow is not the absence of light, but its transformation into differentiated presence.

The pieces operate, therefore, on two simultaneous registers: as objects that carry the historical form of the cobogó, making visible a genealogy that is often simplified; and as material arguments for the central theoretical hypothesis - that the void is not absence, but structure. Ceramics, in this context, does not preserve memory as a written document. It reenacts it through tension, failure, shadow and passage.

## 5. Discussion

The findings of this research suggest that the relationship between cobogó and muxarabi should be treated as a critical affinity, not as a closed genealogy. The initial text risked transforming a poetic hypothesis into an overly strong historical statement. A more careful review of the literature requires a reformulation of the argument: the cobogó is not proof of direct descent, but an indication of an architectural and climatic recurrence that crosses different cultures and material regimes.

This reformulation makes the research more consistent. Colonial erasure does not need to operate as a simple line between origin, prohibition and return. It can function as a system of translations: certain knowledges are disqualified when they belong to the popular, the domestic, the Arab, the colonial or the non-European; later, they may be revalued when they reappear in concrete, modulation, patent, male authorship and national modernism.

The central contradiction of the cobogó is precisely this: it is a legitimate Brazilian invention and, at the same time, participates in a much older and transcultural architectural logic. To affirm only one of these dimensions impoverishes the object. If it is seen only as modern invention, its insertion in a long history of permeable walls is lost. If it is seen only as the return of the muxarabi, its industrial, Recife-based and modern specificity is erased.

Another contribution of the revision is to update the environmental relevance of the theme. The cobogó is no longer only an object of architectural history or decoration: in a context of climate crisis, heat islands and dependence on air-conditioning, it reappears as a strategy of passive cooling. Research on favelas and vulnerable housing indicates that perforated elements can improve ventilation, shading and thermal comfort, although their use depends on security, solar orientation, privacy, maintenance and urban context.

The research presents limits that must be acknowledged. The historical scope remains centred on Brazil and on the Arab-Portuguese-Brazilian trajectory, leaving aside analogous versions of the element in India, North Africa, the Mediterranean and Southeast Asia. The artistic practice is still in process and its reception in exhibition contexts has not yet been analysed. The concept of coloniality of dwelling remains a proposal under construction, which needs to enter into deeper dialogue with contemporary Brazilian researchers in architecture, race, heritage, environmental comfort and climate justice.

## 6. Concluding remarks

This article has argued that the cobogó and the muxarabi should not be reduced either to a direct origin or to a casual resemblance. Between them there is a formal, climatic and political affinity: both make visible an intelligence of perforation, filtering, shade and passage. This intelligence was historically devalued when associated with

repertoires considered backward, colonial or Oriental, and could reappear when translated into the language of industry, concrete and national modernity.

To account for this process, the article has proposed the concept of coloniality of dwelling, articulating contributions from decolonial studies, material culture, phenomenology of space and research on environmental comfort. This theoretical crossroads suggests that architecture does not merely reflect power relations: it produces and reproduces them at the scale of the body, the facade, the climate, the shadow and the air.

The ceramic artistic research that accompanies this theoretical journey does not illustrate the conclusions; it formulates them differently. The pieces pose questions that the text cannot answer alone: what does it mean to support without closing? What does it mean to build with passage? What does clay remember that concrete forgot? The void these pieces carry is not decorative: it is structural, historical and open.

This research is ongoing. What is written here is a provisional map of a territory that still needs to be traversed - in other geographical directions, in other theoretical dialogues, in other pieces. What does the void let through? Air, light, shadow, body, gaze. But also: memory, erasure, return, history. The pieces and this text do not close that question; they keep it open.

## References


APPADURAI, Arjun (ed.). The Social Life of Things: Commodities in Cultural Perspective. Cambridge: Cambridge University Press, 1986.

BACHELARD, Gaston. The Poetics of Space. Boston: Beacon Press, 1994 [1957].

EGBERS, Vera; KAMLEITHNER, Christa; SEZER, Özge; SKEDZUHN-SAFIR, Alexandra (eds.). Architectures of Colonialism: Constructed Histories, Conflicting Memories. Basel: Birkhäuser, 2024.

INGOLD, Tim. Bringing things back to life: creative entanglements in a world of materials. Horizontes Antropológicos, Porto Alegre, v. 18, n. 37, pp. 25-44, 2012.

INGOLD, Tim. Making: Anthropology, Archaeology, Art and Architecture. London: Routledge, 2013.

MERLEAU-PONTY, Maurice. Phenomenology of Perception. London: Routledge, 2002 [1945].

MIGNOLO, Walter D. Local Histories / Global Designs: Coloniality, Subaltern Knowledges and Border Thinking. Princeton: Princeton University Press, 2000.

MIGNOLO, Walter D. The Darker Side of Western Modernity: Global Futures, Decolonial Options. Durham: Duke University Press, 2011.

MOASSAB, Andréia; NAME, Leo. Por um ensino insurgente em arquitetura e urbanismo. Foz do Iguaçu: EDUNILA, 2020.

NAME, Leo; ZAMBUZZI, M. Inconclusive notes on race, architecture and the coloniality of material and immaterial heritage. Epistemologias do Sul, v. 3, n. 1, pp. 118-140, 2019.

PINTO, Estêvão. Muxarabis e balcões. In: Arquitetura Civil II. Rio de Janeiro: IPHAN, 1975.

QUIJANO, Aníbal. Coloniality of power, Eurocentrism and Latin America. Nepantla: Views from South, v. 1, n. 3, pp. 533-580, 2000.

SANTOS, Vanessa Bernardes de Oliveira. Cobogó brise: cobogó com abertura regulável. 2018. Undergraduate thesis (Industrial Design - Product Design) - Escola de Belas Artes, Universidade Federal do Rio de Janeiro, Rio de Janeiro, 2018.

STEFANO, Thaís M.; FIGUEIREDO, Luciana M. B. O uso do cobogó com propósito de melhorar o desempenho térmico em moradias de favelas: análise da literatura. In: ENTAC 2024 - A Construção Civil na Era Digital: pesquisa, inovação e capacitação profissional. Proceedings..., 2024.

VETTORAZZI, Egon; SACHT, Helenice Maria; PUNHAGUI, Katia R. G.; TEIXEIRA, Patricia S. Contribuições da arquitetura árabe para os elementos de controle solar da arquitetura brasileira. Arquitextos, São Paulo, year 23, n. 271.00, Vitruvius, Dec. 2022.